

Audit Trail — Case 11 (reduced work-zone speed): Shoulder closure on divided highway

Field	Value
MUTCD typical application	TA-3
CDOT standard sheet	S-630-1
Case	Case 11 (reduced work-zone speed): Shoulder closure on divided highway
Taper length	150 ft (L/3 (shoulder taper))
Buffer space	360 ft
Device spacing — taper	45 ft
Device spacing — tangent	90 ft
Crew steps	14
Routing	shoulder_reduced_speed

Taper Length

Speed 45 mph $\geq$ 45 mph threshold -> using $L = W \times S$

$$L = 10 \times 45 = 450 \text{ ft}$$

$$L/3 = 450 / 3 = 150 \text{ ft}$$

Field	Value
Closure type	shoulder
Offset (shoulder width)	10 ft
Required taper (L/3 (shoulder taper))	150 ft

Source: MUTCD 11th Ed. Sec 6B.08, Table 6B-3. Shoulder closures use L/3 per Sec 6B.08 (Table 6B-3).

CDOT reference: CDOT S-630-1 Case 11 (right-shoulder closure on divided highway, reduced work-zone speed)

Buffer Space

MUTCD Table 6B-2: 360 ft (CDOT supplement silent at this speed)

Source: MUTCD 11th Ed. Sec 6B.06, Table 6B-2 (stopping sight distance)

Channelizing Device Spacing

45 mph -> 45 ft spacing (MUTCD Sec 6K.01: spacing equals speed in feet)

45 mph -> 90 ft spacing (MUTCD Sec 6K.01: spacing equals 2x speed in feet)

$L/3 = 150 \text{ ft} / 45 \text{ ft max spacing} \rightarrow 5 \text{ drums at } 37.5 \text{ ft intervals}$

$5000 \text{ ft} / 90 \text{ ft max spacing} \rightarrow 57 \text{ cones at } 89.3 \text{ ft intervals}$

Downstream taper: 50 ft at ~25 ft spacing -> 2 cones (MUTCD Sec 6B.08: 50-100 ft downstream taper); 57 tangent + 2 downstream = 59 cones total

Source: MUTCD 11th Ed. Sec 6K.01

Advance Warning Signs

Speed 45 mph, road_type = 'urban_high' -> Table 6B-1 category: urban_high

A = 350 ft, B = 350 ft, C = 350 ft

Position	Code	Station (ft)	Distance from Taper (ft)
Advance (S1-1)	S1-1	7,060	1,550 upstream
C (furthest)	W20-1	6,560	1,050 upstream
W3-5 step 1	W3-5(30)	6,560	1,050 upstream
B (middle)	W20-2	6,210	700 upstream
W3-5 step 2	W3-5(25)	6,030	520 upstream
A (nearest)	W21-5aR	5,860	350 upstream

Source: MUTCD 11th Ed. Table 6B-1

Colorado Requirements (CDOT S-630-1)

Check	Result	Citation	Detail
Signs on both sides of divided highway	FAIL	CDOT S-630-1 (July 2026) Sheet 2, General Note 8	Required: True. Signs placed: 20 left, 22 right.
G20-5P Work Zone signs every 2,640 ft	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 4	Zone length: 6,560 ft. Required: 3. Placed: 5.
Speed reduction <= 15 mph per sign installation	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 3	Work-zone speed reduced 45 -> 25 mph (Delta 20 mph). Required: 2 stepped sign installations per CDOT S-630-1 Sheet 2 General Note 3 (max 15 mph per sign). Placed: 2.
Flagger station lighting 500W @ 8 ft	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 22	Not applicable (no flaggers).

AADT threshold for mobile operations (<= 2,000) — Not applicable (not a mobile operation). (CDOT S-630-1 (July 2026) Sheet 23, Case 38 Note 1)

All Colorado checks pass: False

CDOT Case Reference

Case 11 (reduced work-zone speed): Shoulder closure on divided highway

This scenario matches CDOT Standard Plan S-630-1, Case 11: shoulder closure on a divided highway, with a reduced work-zone posted speed (45 -> 25 mph) that does not match the Case 26/27 mandated step-down. Fines Double envelope applies per S-630-1 Sheet 12, Fines Double Signing Notes.

The generated plan follows the same device layout as the reference case with spacing computed for 45 mph.

Routing: shoulder_reduced_speed

Reference:

[https://www.codot.gov/safety/traffic-safety/assets/s-standard-plans/s-630-1/S-630-01%20\(19-Page%20Set\).pdf](https://www.codot.gov/safety/traffic-safety/assets/s-standard-plans/s-630-1/S-630-01%20(19-Page%20Set).pdf)

Geometry Validation

Work zone 5000 ft vs required taper 150 ft (L/3 (shoulder taper)) + buffer 360 ft.

All geometry checks pass: True

Source: MUTCD 11th Ed. Sec 6B.06 (buffer) and Sec 6B.08 (taper)

Corridor Validation

Corridor check not run (no site coordinates supplied).

Site Adjustments

Site-condition flags layered onto the baseline MUTCD/CDOT layout. Each adjustment is traced to its source rule; the rendered plan, device list, and crew narrative reflect every item below.

Rule	Adjustment
MUTCD §6N.12 p. 848 — Work within the Traveled Way at an Intersection (11th Ed.)	No devices added — the cross-street approach layout is not generated; see the pending-verification disclosure.
MUTCD §6N.16 p. 851 — Interchanges (11th Ed.)	No devices added — the per-ramp interchange layout is not generated; see the pending-verification disclosure.
MUTCD §6C.02 — pedestrian considerations in work zones	Added 4 Type III barricades (sidewalk closure points) and 2 R9-9 SIDEWALK CLOSED signs at the upstream and downstream ends of the work zone.
MUTCD §7B.08 — school zone signing in proximity to work zones	Added 2 S1-1 SCHOOL signs upstream of the advance warning signs (station 7,060 ft, both sides).

Flagger Control

Not applicable for this scenario.

Source: MUTCD 11th Ed. Chapter 6D (Flagger Control) and Chapter 6E (One-Lane, Two-Way Traffic Control).

Fines Double Envelope

Citation: CDOT S-630-1 (July 2026) Sheet 12, Fines Double Signing Notes

Field	Value
R2-10 station	5500 ft
R2-11 station	-500 ft
Envelope length	6000 ft
Assemblies	3
Entrance R2-1	SPEED LIMIT 25 @ 4166.67 ft
Downstream R2-1	SPEED LIMIT 45 @ -1000 ft

Citation	Operational rule
S-630-1 Sheet 12, Note 1	Install Fines Double signs no more than 4 hours before the start of the work day. Do not leave signs up overnight when work is not active.
S-630-1 Sheet 12, Note 2	Remove or cover Fines Double signs when work concludes; doubled fines apply only when workers or work activity are present in the zone.
S-630-1 Sheet 12, Note 3	Relocate the R2-10/R2-11 envelope to follow the actual work area as the project progresses; do not leave Fines Double signs in place beyond the active work zone.
S-630-1 Sheet 12, Note 4	Maintain a 250 ft minimum spacing between Fines Double signs (R2-10, R2-11, G20-5P/R2-6P assemblies) and other warning or regulatory signs. Engineer may adjust placement to satisfy this minimum.

Source: CDOT S-630-1 Standard Plan, Sheet 12

Pending Verification

2 reference(s) pending verification. An adjacent intersection is flagged, but Conestruct does not generate the cross-street approach layout or evaluate traffic-signal operation, and adds no devices for it. Cross-street control design and the signal-operation review (MUTCD §6N.12, 11th Ed. p. 848; Part 4 for signal-head visibility) remain with the traffic control supervisor. Intersection support is tracked at issue #117; corner-quadrant work at issue #128. Tracking: <https://github.com/rtmakatura/conestruct/issues/117>

- [intersection_layout_not_generated] An adjacent intersection is flagged, but Conestruct does not generate the cross-street approach layout or evaluate traffic-signal operation, and adds no devices for it. Cross-street control design and the signal-operation review (MUTCD §6N.12, 11th Ed. p. 848; Part 4 for signal-head visibility) remain with the traffic control supervisor. Intersection support is tracked at issue #117; corner-quadrant work at issue #128. (<https://github.com/rtmakatura/conestruct/issues/117>)
- [interchange_layout_not_generated] An adjacent interchange is flagged, but Conestruct does not generate the per-ramp layout and adds no devices for it. Ramp-specific control design and the ramp-access and coordination review (MUTCD §6N.16, 11th Ed. p. 851) remain with the traffic control supervisor. Tracked at issue #117. (<https://github.com/rtmakatura/conestruct/issues/117>)